HAMZA BENAMMI

SENIOR SOFTWARE ENGINEER | BACKEND, AWS & CLOUD-NATIVE SYSTEMS

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PROFESSIONAL PROFILE

Senior Software Engineer with 8+ years of experience designing and delivering SaaS products, backend platforms, enterprise applications, data-automation tools and cloud-native systems. Advanced hands-on expertise in Node.js, TypeScript, Python, PHP and AWS serverless architecture, with ownership of business logic, REST APIs, data models, security middleware, CI/CD and production troubleshooting. Led backend engineering and delivery for Vially and Stryve, previously contributed to a large multi-module ERP, and built Python desktop software for automated web data collection and vehicle-pricing workflows. Strong engineering-quality background across unit, API, integration, end-to-end and manual testing, with additional experience in AI/RAG, Docker, Kubernetes and developer tooling.

CORE EXPERTISE

Backend Architecture & APIs
Node.js, Laravel and business logic

AWS & Infrastructure as Code
Serverless systems and CloudFormation

Technical Leadership
Sprint planning, estimation and delivery

Data & Security
DynamoDB, PostgreSQL, RBAC and permissions

Quality Engineering
Unit, API, integration and E2E testing

Automation & Developer Tooling
CLI packages, scraping and CI/CD

Enterprise Applications
ERP systems, Vue.js and Livewire

Production Engineering
Observability, defect triage and root cause

AI Engineering
Bedrock, RAG, fine-tuning and Ollama

PROFESSIONAL EXPERIENCE

INVENEO | AWS Engineer (Senior Backend & Technical Lead Responsibilities)

Aug 2025 - Present

Remote | Morocco

- Lead backend engineering and business logic for Vially, structuring Node.js services and AWS serverless workflows for accessibility scanning, monitoring, reporting and analytics.
- Define solution architecture, REST API contracts, data flows and integration requirements; coordinate with frontend engineers and manually validate that business-critical workflows are implemented correctly.
- Built and evolved a website accessibility audit engine using Selenium, Playwright and Node.js, including authenticated sessions, configurable browser/device behaviour, custom scripts, screenshots, multi-page processing and structured reporting.
- Create reusable Node.js command-line packages that automate audits and expose consistent commands for CI/CD pipelines and other automation systems.
- Design DynamoDB schemas and access patterns, provision infrastructure through CloudFormation and manual AWS configuration, and work with PostgreSQL where relational modelling is required.
- Implement and review authentication, role-based access control and permissions middleware for multi-tenant SaaS products.
- Plan sprints and timelines in Jira, break requirements into deliverable tickets, prioritise work, review technical implementation and support defect triage and root-cause analysis.
- Use Cypress, Jest, Vitest, Selenium, Playwright, API tests and manual validation to protect releases and verify backend-to-frontend integration.

STRYVE | AWS Engineer (Backend Platform & Team Lead Responsibilities)

Nov 2022 - Aug 2025

Remote | Morocco

- Led backend engine and business-logic delivery for an enterprise accessibility SaaS platform, defining service boundaries, shared modules, API behaviour and deployment patterns across development, staging, pre-production and production.
- Built and maintained cloud-native services with Node.js, TypeScript, AWS Lambda, API Gateway, S3, DynamoDB, CloudFront, CloudWatch, CodePipeline and CodeBuild.
- Developed browser automation, website scanning and data-extraction workflows using Selenium, Puppeteer, Playwright, Node.js and Python.
- Implemented automated UI and integration tests with Cypress, unit tests with Jest and Vitest, and API/regression testing; also performed manual acceptance testing and verified frontend delivery against business requirements.
- Designed monitoring, alerting, media-asset reporting, scan analytics and issue-lifecycle workflows, including closing resolved findings and reopening recurring issues across repeated scans.
- Built service-oriented and multi-service architectures using Docker and Kubernetes, with CI/CD practices for repeatable delivery.
- Implemented and validated Stripe billing flows including plan selection, upgrades, add-ons and tax previews, combining backend business logic with reliable frontend integration.
- Collaborated with product, frontend, DevOps and business stakeholders; planned Jira sprints, estimated timelines, reviewed implementation and investigated production issues through logs and cloud data.

PROFESSIONAL EXPERIENCE - CONTINUED

FREELANCE | Software Engineer - Python Data Automation
Aug 2022 - Oct 2022

Morocco

- Designed and delivered a Python desktop application that automated web data collection, cleaned and normalised extracted information, and transformed it into structured data used by a vehicle-pricing website.
- Built the end-to-end workflow covering desktop UI, browser and scraping automation, data validation, transformation and export, replacing a manual data-preparation process with a reusable software tool.

PROFONDE SOLUTION | Full-stack Developer - Enterprise ERP
Oct 2021 - Jul 2022

Marrakesh, Morocco

- Contributed to a large, multi-module ERP platform supporting complex operational workflows, business rules and high-volume structured data across the organisation.
- Developed backend modules and business logic with PHP and Laravel, and delivered interactive application features using Vue.js and Livewire.
- Designed and queried PostgreSQL data models, integrated application modules and maintained reliable data flows across frontend, backend and database layers.
- Worked in an Agile/Scrum environment using GitLab and Trello for version control, planning, issue tracking and team collaboration; investigated defects and delivered features from requirement through release.

FREELANCE | Web Developer
Oct 2018 - Oct 2021

Morocco

- Delivered websites and web applications for clients, covering requirements discovery, solution structure, frontend and backend implementation, testing, deployment and ongoing support.
- Built practical experience in independent delivery, customer communication, troubleshooting and maintaining production websites across different business needs.

SELECTED ENGINEERING CONTRIBUTIONS

Enterprise Accessibility Audit Platform | Node.js | Selenium | Playwright | AWS | WCAG

Designed an automated engine that scans complex websites, supports authenticated sessions and device simulation, runs custom accessibility criteria, captures evidence and produces structured reports for repeated audits and monitoring.

Reusable Audit CLI and Automation Packages | Node.js CLI | CI/CD | Developer Tooling

Created reusable command-line packages that standardise audit execution and integrate with pipelines, scheduled jobs and other engineering systems.

AI, RAG and Local Agent Systems | Amazon Bedrock | RAG | Ollama

Built AI-assisted systems using Bedrock and local Ollama models, including retrieval-augmented generation, fine-tuning experiments, web-aware assistants, code generation and document workflows.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Node.js, Python, PHP and SQL; earlier experience with Java, C#, ASP.NET and ADO.NET

Backend & Frameworks: REST APIs, Express, Laravel, service-layer architecture, business logic, middleware, authentication, RBAC, permissions and multi-tenant SaaS

Cloud & DevOps: AWS Lambda, API Gateway, S3, DynamoDB, CloudFront, CloudWatch, CodePipeline, CodeBuild, CloudFormation, CI/CD, GitHub and GitLab

Databases: PostgreSQL, DynamoDB, MySQL, relational modelling, access-pattern design, data validation and migrations

Testing & Automation: Cypress, Jest, Vitest, Selenium WebDriver, Playwright, Puppeteer, Postman, unit, API, integration, end-to-end, regression and manual testing

Frontend & Enterprise UI: Vue.js, Livewire, React, TypeScript, API integration and implementation validation

Platforms & AI: Docker, Kubernetes, Linux, Amazon Bedrock, RAG, fine-tuning, Ollama, prompt engineering and local AI agents

Delivery & Collaboration: Agile/Scrum, sprint planning, estimation, Jira, Trello, code review, technical documentation and cross-functional delivery

EDUCATION

ISTA NTIC | Diploma in Software Development
2016 - 2018

Marrakesh, Morocco

ISIM | Diploma in IT Maintenance, Support & Networks
2013 - 2015

Marrakesh, Morocco

ADDITIONAL EXPERIENCE & LANGUAGES

- Faculté des Sciences Semlalia Marrakech - Full-stack Web Developer, Mar 2018 - Apr 2018: developed an e-learning platform.
- ISGA Campus Marrakech - Trainee, Mar 2015 - Apr 2015: installed and administered an OwnCloud server and a local network.

Arabic | French | English